Calibration Plots

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Introduction

A calibration plot (reliability diagram) plots predicted probability against observed frequency. A well-calibrated classifier's curve lies on the diagonal: when it predicts 0.7, events occur 70 % of the time. Poor calibration means probabilities cannot be trusted for decision thresholds or risk communication.

Prerequisites

Classification; predicted probabilities; binning.

Theory

Bin predictions by predicted probability; compute the observed frequency in each bin. Plot bin midpoints vs observed frequencies. A perfect classifier follows $y = x$; curves below indicate over-confidence, curves above indicate under-confidence.

Summary metrics: **Brier score** $= \mathbb{E}[(p-y)^2]$, **Expected Calibration Error (ECE)** $= \sum_k w_k |\text{acc}_k - \text{conf}_k|$.

Assumptions

Predictions are valid probabilities in $[0, 1]$; sufficient data per bin (10-20 bins for $n > 1000$).

R Implementation

```r
library(probably); library(yardstick); library(dplyr); library(tibble); library(ggplot2)

set.seed(2026)
n <- 1000
# Simulate mis-calibrated predictions
true_y <- rbinom(n, 1, 0.4)
raw_p  <- plogis(-1 + 2 * true_y + rnorm(n, 0, 1.5))
# Inflate confidence to produce over-confident predictions
p_overconf <- raw_p^0.6
p_overconf <- pmin(pmax(p_overconf, 0.02), 0.98)

df <- tibble(truth = factor(true_y, levels = c(1, 0)),
             pred  = p_overconf)

cal_df <- cal_plot_breaks(df, truth, pred, num_breaks = 10)

# Manual reliability data + plot
df$bin <- cut(df$pred, seq(0, 1, 0.1), include.lowest = TRUE)
reliability <- df %>%
  group_by(bin) %>%
  summarise(mean_pred = mean(pred),
            obs_freq  = mean(as.integer(truth == "1")),
```

```r
          n = n())

ggplot(reliability, aes(mean_pred, obs_freq)) +
  geom_point(size = 2, colour = "#2A9D8F") +
  geom_line() +
  geom_abline(linetype = 2) +
  labs(x = "Predicted probability", y = "Observed frequency",
       title = "Reliability diagram (simulated over-confident model)") +
  theme_minimal()

mean((df$pred - as.integer(df$truth == "1"))^2)   # Brier score
```

Output & Results

A reliability diagram (scatter of bin-means) and the Brier score. The simulated over-confident model lies above the diagonal at low predicted probabilities.

Interpretation

"The classifier over-predicts at low probabilities and under-predicts at high probabilities; Brier score 0.22 indicates moderate calibration error. Platt or isotonic recalibration on a validation set would flatten the curve toward the diagonal."

Practical Tips

- Tree ensembles (RF, XGBoost) are usually mis-calibrated; apply Platt scaling or isotonic regression on a validation set.
- Neural networks with modern training (high-capacity, low label smoothing) often over-confident; temperature scaling fixes it.
- Use stratified bins for reliable plots when probabilities concentrate near 0 or 1.
- Report Brier score and ECE alongside ROC AUC; AUC is rank-based and does not reveal calibration.
- For clinical applications, calibration matters more than discrimination; always verify on deployment-era data.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis